

Berfin Celik

ML/AI & Backend Engineer

Leeds, UK | Flexible on location | Graduate Route visa (until 17 Nov 2027) | berfincelik0@outlook.com

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Software engineer working across backend systems and machine learning: 2+ years of production .NET Core experience and an MSc in Computer Science & Artificial Intelligence (Distinction, University of Bradford). Builds and ships REST APIs, microservices, NLP pipelines, LLM fine-tuning (LoRA/QLoRA), and RAG systems.

EDUCATION

University of Bradford - MSc Computer Science & Artificial Intelligence | **Distinction** 2024 - 2025

Dissertation: Designed and evaluated an end-to-end NLP pipeline for multidialectal language understanding in low-resource languages (Turkish, Kurmanji, Zazaki), combining ASR, topic modelling, and knowledge graph construction. Fine-tuned Whisper and Wav2Vec2 on a custom corpus; implemented BERTopic, KeyBERT/YAKE, and NetworkX for semantic knowledge graphs. Evaluated with WER, CER, BLEU, and topic coherence, with human-in-the-loop validation by native speakers. ([PDF](#)) ([Code](#))

Bahcesehir University - BSc Computer Engineering 2016 - 2021

EXPERIENCE

Junior .NET Developer - Path Product & Software House Istanbul, Turkey Dec 2023 - May 2024

- Optimised critical back-end services in C# and .NET Core for high-load applications, improving API response times by ~35% through performance tuning and algorithmic improvements.
- Built microservices based on Onion Architecture, improving modularity and maintainability; integrated Redis distributed caching to reduce database load and improve retrieval speeds.
- Containerised applications with Docker and collaborated within Agile/Scrum teams, enhancing deployment consistency and overall system stability.
- Contributed to a school portal system from initial architecture through backend services, caching, and deployment.

Software Engineer - Argevi Software & Automation Istanbul, Turkey Mar 2022 - Jun 2023

- Optimised back-end systems and database schemas through query tuning, reducing average query time by ~40% on high-traffic applications.
- Engineered and deployed a Project Management Tool (ASP.NET Core, React) and an E-commerce Platform (.NET Core, Angular) with secure payment gateways, Redis caching, and Docker containerisation.
- Implemented OPC industrial-protocol communication from scratch and delivered it to production; used Kubernetes for orchestration and RabbitMQ for message brokering.

IT Intern - Turkuvaz Media Group Istanbul, Turkey Dec 2020 - Mar 2021

- Tested updated application components and monitored database performance ahead of production releases.
- Assisted in optimising MySQL queries, improving data retrieval speeds and overall system efficiency.

Student Assistant - Bahcesehir University Engineering Faculty Istanbul, Turkey Nov 2019 - Mar 2020

PROJECTS

University Chatbot — LLM Fine-Tuning & RAG - LLaMA 2, DeepSeek, QLoRA, FAISS, Hugging Face ([GitHub](#))

- Scraped SharePoint content into 600+ QA pairs; fine-tuned LLaMA 2 and DeepSeek with QLoRA and implemented FAISS-based RAG retrieval, improving answer relevance and reducing hallucinations.
- Benchmarked four model variants on retrieval and generation quality; deployed via Hugging Face Spaces.

AI-Powered Deepfake Voice Detection (team project) - CNN, Whisper, LoRA, Quantisation, FastAPI, Docker ([GitHub](#))

- Engineered a hybrid CNN + Whisper pipeline for synthetic audio detection using MFCC features and transformer embeddings; applied LoRA fine-tuning and quantisation for efficient multilingual inference.
- Trained and evaluated on 220k+ real and synthetic audio samples; deployed as a FastAPI service containerised with Docker.

Dialectal Word Mining & Knowledge Graph - Whisper, Wav2Vec2, NetworkX, Python ([GitHub](#))

- Mined dialect vocabulary across Turkish, Kurmanji, and Zazaki via ASR; built co-occurrence knowledge graphs with NetworkX.

PROFESSIONAL DEVELOPMENT

- 2025 – present: continuing ML specialisation — completed IBM's 'Generative AI Language Modeling with Transformers' and 'Introduction to Neural Networks and PyTorch' (Coursera, 2026).
- Maintaining an open-source ML portfolio and portfolio site (github.com/berfinito · berfinito.github.io).

SKILLS

Programming: Python, C#, .NET Core / ASP.NET, SQL, Java, C, C++, JavaScript

Backend: ASP.NET Core / .NET, FastAPI, REST APIs, Microservices (Onion Architecture), Redis caching, RabbitMQ

ML & Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, LLM Fine-tuning (LoRA, QLoRA), NLP (ASR, RAG), Whisper, Wav2Vec2, Model Quantisation, FAISS, BERTopic, NetworkX, Gradio

Cloud, DevOps & MLOps: Docker, Kubernetes, CI/CD (GitHub Actions), GCP, Redis, RabbitMQ

Databases: PostgreSQL, MySQL, SQLite, Redis

Languages: English (Proficient), Turkish (Native)